MCP with Github

> GitHub REST API를 `httpx`로 호출하고 **FastMCP** 도구로 노출하는 과정을 다룹니다.

Github API

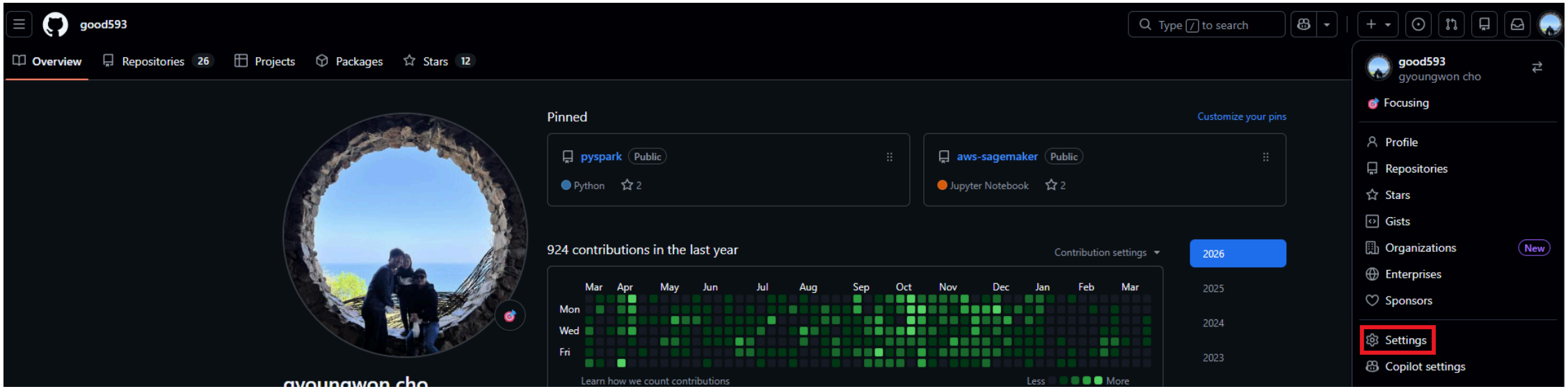
> GitHub REST API를 사용하여 통합을 만들고, 데이터를 검색하고, 워크플로를 자동화합니다.

GitHub REST API 핵심

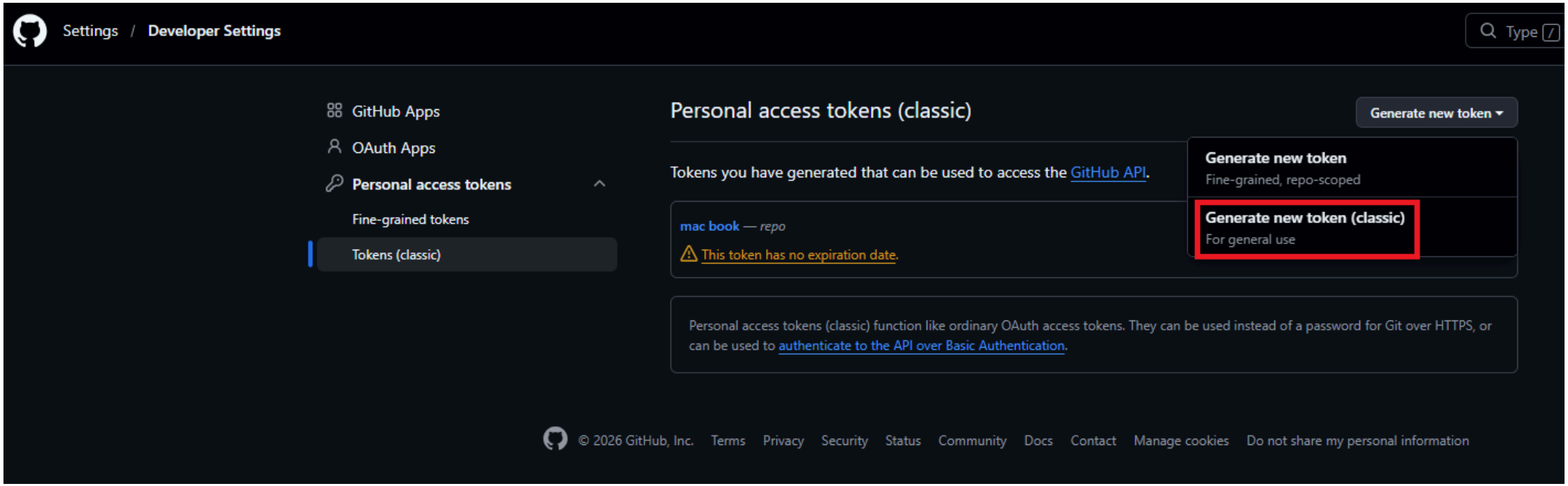
항목	내용
Base URL	`https://api.github.com`
권장 헤더	`Accept: application/vnd.github+json`, `X-GitHub-Api-Version: 2022-11-28`
인증(실습)	`Authorization: Bearer <PAT>`
응답	대부분 JSON. 오류 시 `message` 필드 등 확인

API Token 생성하기

1) Github Login > Setting



2) Setting > Developer Settings > Tokens



3) New personal access token (classic)

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.

Note

mcp-server

What's this token for?

Expiration

7 days (Mar 28, 2026)

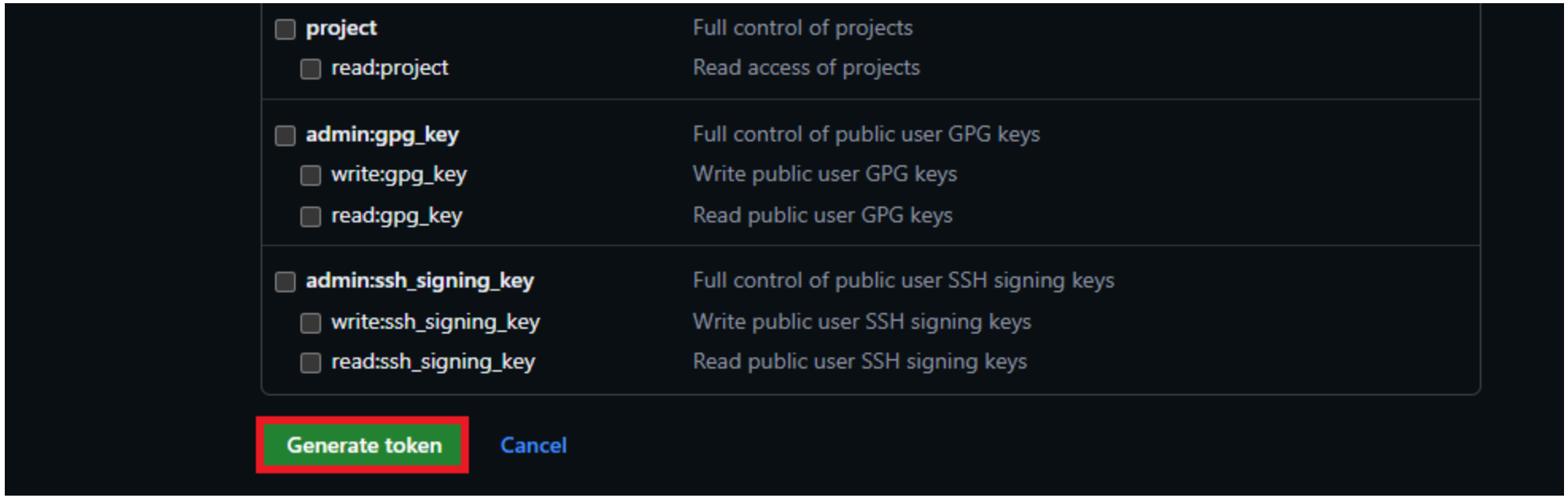
The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. Read more about OAuth scopes.

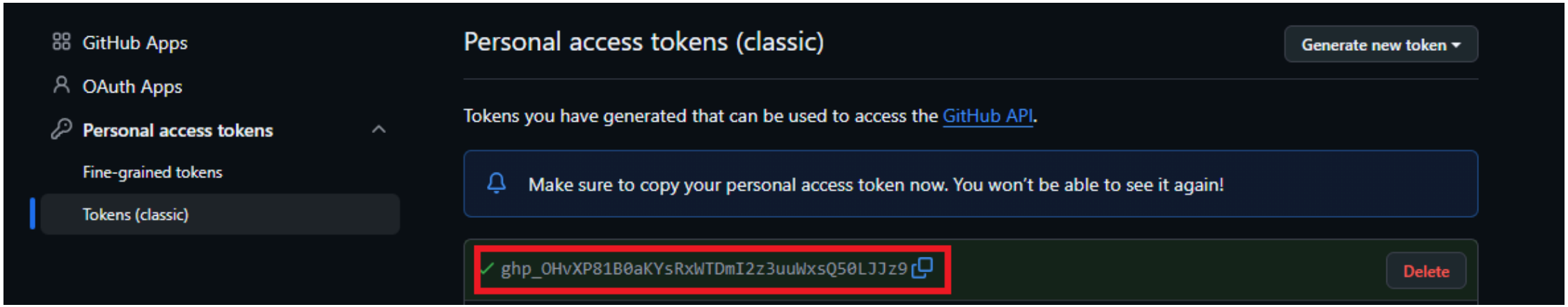
☑ **repo**	Full control of private repositories
☑ repo:status	Access commit status
☑ repo_deployment	Access deployment status
☑ public_repo	Access public repositories
☑ repo:invite	Access repository invitations
☑ security_events	Read and write security events
☐ **workflow**	Update GitHub Action workflows

Token 생성하기

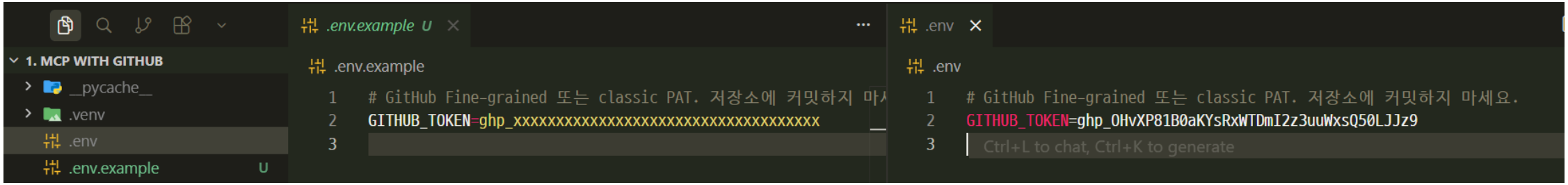

project
Full control of projects
read:project
Read access of projects
admin:gpg_key
Full control of public user GPG keys
write:gpg_key
Write public user GPG keys
read:gpg_key
Read public user GPG keys
admin:ssh_signing_key
Full control of public user SSH signing keys
write:ssh_signing_key
Write public user SSH signing keys
read:ssh_signing_key
Read public user SSH signing keys
Generate token
Cancel

4) .env 파일에 Token 복사하기

생성된 Token 복사하기

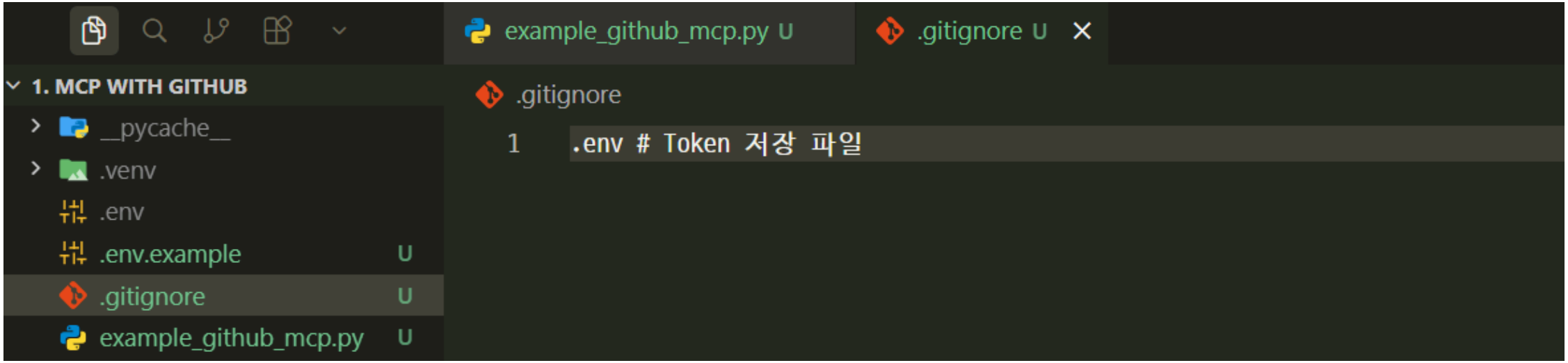


.env.example을 이용해서 .env 파일 생성



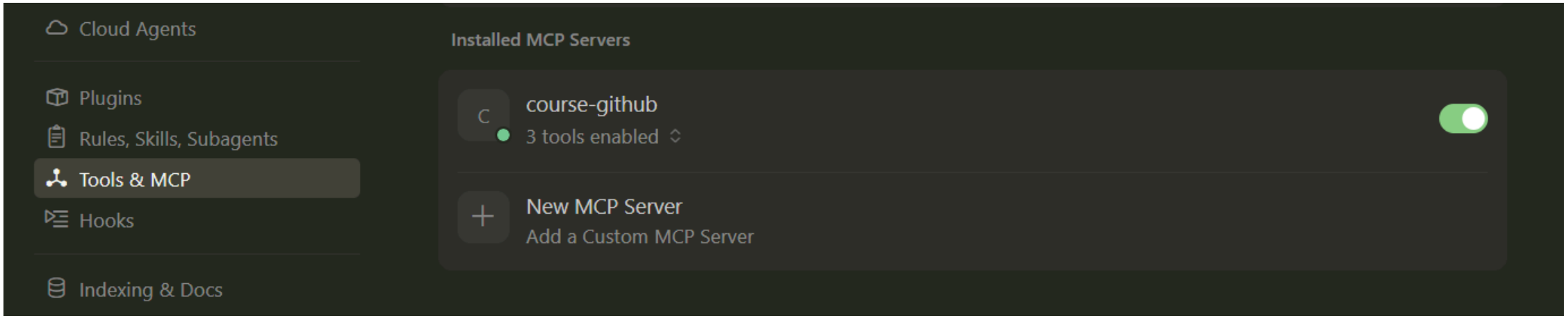
5) .gitignore

> 프로젝트 루트에 `.env` 파일을 두고 **절대 커밋하지 않는다**.



MCP를 Cursor에 적용하기

```
"course-github": {
  "command": "C:/develop/github/course_LLM/6. MCP/3. MCP with API/1. MCP with GitHub/.venv/Scripts/python.exe",
  "args": ["C:/develop/github/course_LLM/6. MCP/3. MCP with API/1. MCP with GitHub/example_github_mcp.py"],
  "cwd": "C:/develop/github/course_LLM/6. MCP/3. MCP with API/1. MCP with GitHub"
}
```



테스트 > course-github를 이용해서 course로 시작하는 저장소 조회해줘.

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Explored 1 file

MCP `search_repositories` 로 검색합니다. GitHub는 이름 '접두어' 전용 연산자가 없어 `in:name` 과 `course` 로 검색합니다.

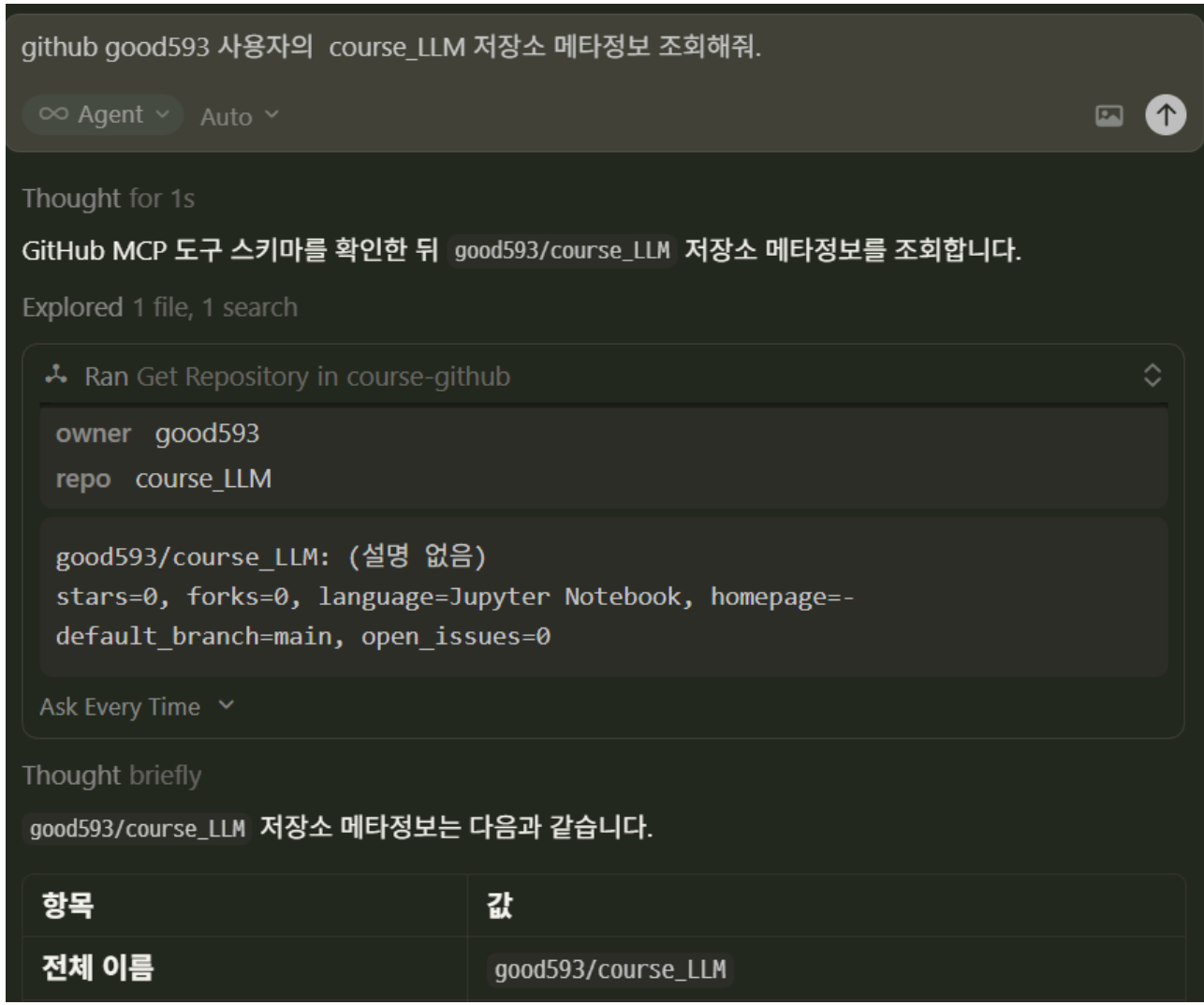
Ran Search Repositories in course-github

Ask Every Time

Thought for 3s

`course-github` (MCP 서버 `user-course-github`)의 `search_repositories` 도구로 `course in:name` 쿼리를 실행했습니다. GitHub 저장소 검색은 이름 **접두어만** 고르는 전용 문법이 없어서, 이름에 `course` 가 들어가는 저장소를 찾은 뒤 그중에서 **저장소 이름이** `course` **로 시작하는 것**만 정리했습니다.

검색 결과 (이름이 `course` / `Course` 로 시작)

저장소	별점	설명 요약
huggingface/course	★3782	Hugging Face Transformers 강의

테스트 > github good593 사용자의 course_LLM 저장소 메타정보 조회해줘.